

Conduction, Convection, & Radiation

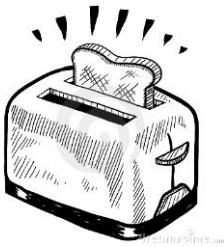
Define the 3 words below. Then fill in the blanks with the word that best represents what is happening.

Define conduction: how we heat things through objects

Define convection: how warm/cold air separates

Define radiation: heat or light coming off objects

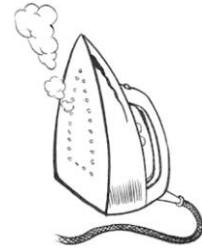
PART I:



1. radiation



2. convection



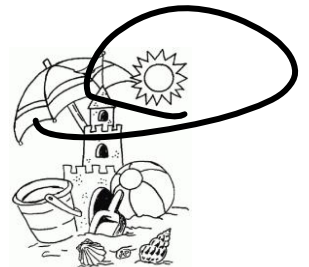
3. convection



4. radiation



5. conduction



6. radiation



7. radiation



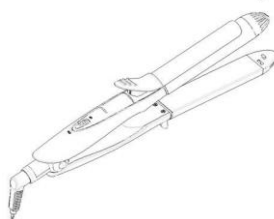
8. radiation



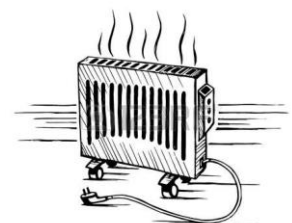
9. convection



10. conduction



11. conduction



12. radiation

PART II:

1. Hot coffee is stirred with a spoon, the spoon gets hot due to conduction.
2. A chair is placed several feet from a fire in a fireplace. The fireplace has a glass screen. The side of the chair facing the fireplace gets warm because of radiation.
3. A certain type of decorative lamp contains colored liquids (aka: a Lava Lamp!). These liquids form globs that break off and rise to the top of the liquid. The globs rise due to convection.
4. Near the ceiling of a room the air is warmer. The warm air rises because of convection.
5. A student holds the back of his hand near an iron to see if it is hot. Heat is transferred to his hand by radiation.
6. A certain type of stainless steel cookware has a layer of copper applied to the bottom to help it heat evenly. The copper transfers heat to the pan by conduction.
7. In a swimming pool, the water near the surface is slightly warmer. The warm water rises because of convection.
8. One end of a copper rod is placed in a flame of a Tirril burner. Small pieces of wax placed along the rod melt at progressively larger distance from the flame. Heat is transferred through the rod by conduction.
9. A house burns down. On the house across the street, all of the vinyl siding is twisted and warped by the heat. The heat was transferred across the street by radiation.
10. Warm air over the beach rises while cooler dense air from the ocean rushes in due to convection.
11. The metal skewer gets so hot that you drop your marshmallow in the campfire because of conduction.
12. A huge rock at the state park gets so hot during the day that you can't sit on it from conduction.
13. A fireman feels a door and it is hot from the fire on the other side due to radiation.
14. You are in the top bunk of a bunk bed and you want to turn the air conditioner on while your friend on the bottom bunk is fine is caused by convection.